Project Design Phase-II

Solution Requirements (Functional & Non-Functional)

Date	08 March 2026
Team ID	—
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub-Requirements
FR-1	User Health Data Input	System provides a web form where users enter health parameters such as age group, gender, symptoms, blood pressure levels, medication history, and diet control
FR-2	Data Validation	System validates all input fields and ensures required health parameters are entered before prediction
FR-3	Data Preprocessing	System converts categorical inputs into numerical format and scales features for machine learning prediction
FR-4	Hypertension Risk Prediction	Machine learning model predicts hypertension stage based on processed user input data
FR-5	Risk Level Display	System displays predicted hypertension stage along with risk level indicators
FR-6	Confidence Score	System provides prediction confidence percentage generated by the machine learning model
FR-7	Health Recommendations	System generates medical advice and lifestyle suggestions based on predicted hypertension stage
FR-8	Web Interface Integration	Flask application serves the user interface and handles communication between frontend and machine learning model
FR-9	Result Visualization	Prediction results are displayed clearly on the web interface with color-coded risk levels and recommendations

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Web interface should be simple, intuitive, and easy for users to input health data and understand prediction results
NFR-2	Performance	Prediction response time should be fast (typically less than 2 seconds) to provide instant health risk assessment
NFR-3	Reliability	System should process user inputs without crashing and handle incorrect inputs gracefully
NFR-4	Scalability	Architecture should allow integration of additional machine learning models or more health parameters in future

NFR-5	Availability	System should run on any machine with Python, Flask, and required libraries installed
NFR-6	Security	Input validation ensures that incorrect or malicious data cannot affect the system
NFR-7	Maintainability	Codebase should be modular with separate components for frontend, backend, and machine learning model